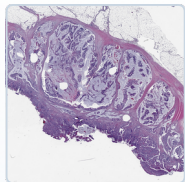


Four independent real-measurement datasets probe what virtual mIF encodes

A model that has only ever seen H&E, tested against four kinds of ground truth.

INPUT



H&E only

No mIF, no IHC
no clinical metadata

EXCLUDED FROM THE INPUT

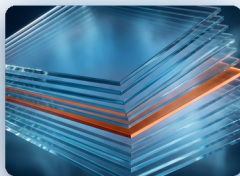
mIF ×

IHC ×

metadata ×

every number in this figure
comes from this one image

VIRTUAL mIF



23 channels

predicted, never measured

GigaTIME

U-Net++ · 9.16 M params

GigaTIME-Flash

DINOv2-small + LoRA · 23.8 M

both reach Dice 0.36 on the
authors' own released sample tiles

tested

FOUR INDEPENDENT REAL MEASUREMENTS

L1

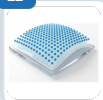


PIXEL HEMIT · colorectal mIHC

pixel · 945 patches · 3 channels

DAPI Dice 0.78 · CK 0.47–0.52 · **CD3 never ranks 1st**

L2



SPOT GSE230424 · thyroid

spot · **4 samples** · 15,489 spots

only **3 of 11 pairs** pass the specificity test

L3



PATIENT TCGA-THCA + RPPA

patient · 222 patients · 175 proteins

paired channels rank **10 / 17 / 21** of 23

L4



CELL ORION-CRC

single cell · 41 patients · 994,729 cells

only **2 of 13 pairs** pass in over half the patients

Specificity = the paired association minus the strongest competing association.